

Time Series Analysis

Seminar 2: ARMA models



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **EODHD**: EOD Historical Data (market-data provider)
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices
- **JB**: Jarque–Bera (test)
- **MA**: Moving Average
- **MAE**: Mean Absolute Error
- **MLE**: Maximum Likelihood Estimation
- **PACF**: Partial AutoCorrelation Function
- **RMSE**: Root Mean Squared Error
- **US**: United States
- **WN**: White Noise

Today's Question and Seminar Structure

- The question of the seminar
  - **how do we choose, estimate, check and use an ARMA model for a stationary series?**
- Results to obtain (nothing is handed in)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C2
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	AR(1): moments and forecasts; AR(2): roots and autocorrelations	Solved, Proposed	A1
A3, A4	MA(1) invertibility; ARMA(1,1) weights and a common factor	Solved, Proposed	A3
A5, A6	Yule–Walker and information criteria; criteria and Ljung–Box degrees of freedom	Solved, Proposed	A5
B1, B2	Box–Jenkins: US real GDP growth; Romanian real GDP growth	Solved, Proposed	B1
B3, B4	AR(1) for daily returns: BET; EUR/RON	Solved, Proposed	B3
C1, C2	does an AR beat simple forecasts of inflation? what is wrong in an AI answer?	Proposed	B1, A1

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in class
- Part A: calculations on paper; Part B: real data in the notebook, each task ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data used

Series	Source	Frequency	Period
US real GDP	statsmodels `macrodata`	quarterly	1959–2009
Real GDP, Romania	Eurostat (namq_10_gdp)	quarterly, seasonally adjusted	2000–2026
BET	EODHD	daily, close	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2005–2026

- Growth rates in %: $400\,\Delta \ln Y_t$ (US, annualised), $100\,\Delta \ln Y_t$ (Romania, quarter on quarter); daily log returns $100\,\Delta \ln P_t$
- 12-month inflation: $100\,(\ln P_t - \ln P_{t-12})$
- In the notebook: `fit_arma(x, p, q)`, `box_jenkins(y, name)`, `b3_returns('bet')`; no account or key is needed

Prerequisites for Today (1/17): the models

- $\varepsilon_t \sim \text{WN}(0, \sigma^2)$: white noise (Seminar 1); **lag operator** L: $LX_t = X_{t-1}$, $L^j X_t = X_{t-j}$
- **AR(p)** (autoregressive): today's value from the last p values

$$X_t = c + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t, \qquad \phi(L) X_t = c + \varepsilon_t$$

  - c: the intercept; ϕ_i: the weights of the past values; $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$: the AR polynomial
- **MA(q)** (moving average): today's value from the last q shocks

$$X_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q} = \mu + \theta(L)\varepsilon_t$$

  - μ: the mean; θ_j: the weights of the past shocks; $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$: the MA polynomial
- **ARMA(p, q)**: both parts; p, q: the orders

$$\phi(L)(X_t - \mu) = \theta(L)\varepsilon_t$$

Prerequisites for Today (2/17): stationarity and invertibility

- **Stationary (causal)**: all roots of $\phi(z) = 0$ lie outside the unit circle, $|z| > 1$

$$X_t = \mu + \sum_{j \geq \mathbf{0}} \psi_j \varepsilon_{t-j}, \qquad \psi_{\mathbf{0}} = 1$$

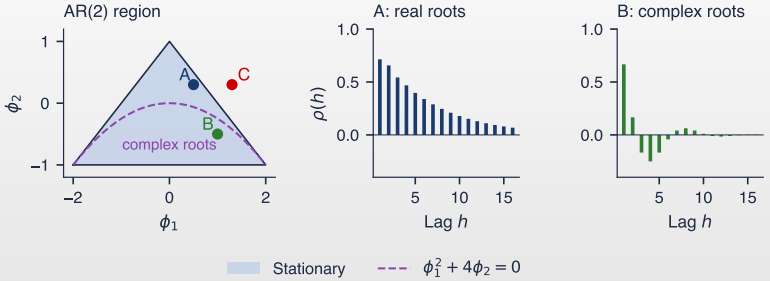
  - ψ_j: the effect on X_t of a shock that occurred j periods earlier (the MA(∞) form)
  - AR(1): $|\phi| < 1$; AR(2): $\phi_{\mathbf{1}} + \phi_{\mathbf{2}} < 1$, $\phi_{\mathbf{2}} - \phi_{\mathbf{1}} < 1$, $|\phi_{\mathbf{2}}| < 1$ (a triangle)
  - a root with $|z| < 1$: the process explodes; a root $z = 1$: a unit root (Seminar 3)

- **Invertible**: all roots of $\theta(z) = 0$ satisfy $|z| > 1$

$$\varepsilon_t = \sum_{j \geq \mathbf{0}} \pi_j (X_{t-j} - \mu), \qquad \pi_{\mathbf{0}} = 1$$

  - the shocks can be recovered from the past values (the AR(∞) form); MA(1): $|\theta| < 1$, $\pi_j = (-\theta)^j$
  - stationarity depends only on $\phi(z)$, invertibility only on $\theta(z)$

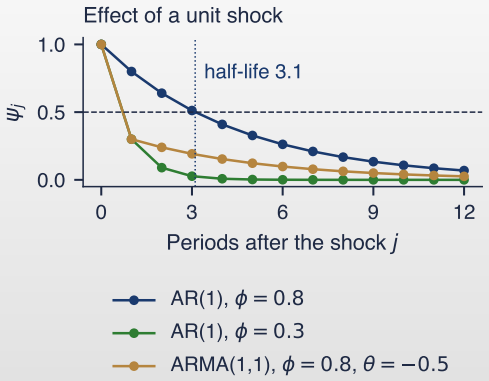
Prerequisites for Today (3/17): the AR(2) triangle



- Left: the pairs (ϕ_1, ϕ_2) inside the triangle give a stationary AR(2); below the dashed parabola the roots are complex
- Three examples
  - A $= (0.5, 0.3)$: real roots 1.17 and 2.84; the ACF decays smoothly
  - B $= (1.0, -0.5)$: complex roots with $|z| = 1.41$; the ACF is a damped wave
  - C $= (1.3, 0.3)$: $\phi_1 + \phi_2 = 1.6 > 1$, a root 0.67 inside the circle: explosive

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Prerequisites for Today (4/17): AR(1): moments and half-life



- **AR(1)** $X_t = c + \phi X_{t-1} + \varepsilon_t$, $|\phi| < 1$

$$\mu = \frac{c}{1-\phi}, \quad \gamma(0) = \frac{\sigma^2}{1-\phi^2}, \quad \rho(h) = \phi^h, \quad \psi_j = \phi^j$$

  - μ: the mean, not the intercept c; $\gamma(0)$: the variance; $\rho(h)$: the autocorrelation at lag h

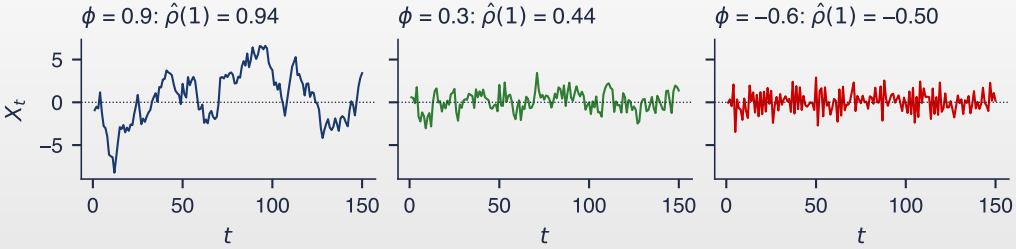
- **Half-life** of a shock: the number of periods after which half of its effect remains

$$\phi^j = 0.5 \iff j = \frac{\ln 0.5}{\ln \phi}$$

  - chart: $\phi = 0.8$ gives $\ln 0.5 / \ln 0.8 = 3.1$ periods; $\phi = 0.3$ gives 0.6
  - the ARMA(1,1) line: $\psi_1 = \phi + \theta$, then each ψ_j is ϕ times the previous one

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Prerequisites for Today (5/17): three AR(1) paths



- Simulated with the same shocks, $T = 150$, $c = 0$; the title gives ϕ and the sample $\hat{\rho}(1)$
- The size and the sign of ϕ
  - $\phi = 0.9$: long waves, slow return to the mean (half-life 6.6 periods)
  - $\phi = 0.3$: close to white noise; $\phi = -0.6$: a zigzag, a high value is usually followed by a low one
  - in each case $\hat{\rho}(1)$ is close to ϕ, since $\rho(1) = \phi$

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Prerequisites for Today (6/17): moments of AR(2), MA(1) and ARMA(1,1)

- **AR(2)**, Yule–Walker equations: each $\rho(h)$ follows the AR recursion

$$\rho(1) = \frac{\phi_1}{1-\phi_2}, \qquad \rho(h) = \phi_1\rho(h-1) + \phi_2\rho(h-2),\ h \geq 2$$

  - complex roots when $\phi_1^2 + 4\phi_2 < 0$: the ACF is a damped wave; $\psi_j = \phi_1\psi_{j-1} + \phi_2\psi_{j-2}$, $\psi_0 = 1$, $\psi_1 = \phi_1$
- **MA(1)** $X_t = \mu + \varepsilon_t + \theta\varepsilon_{t-1}$

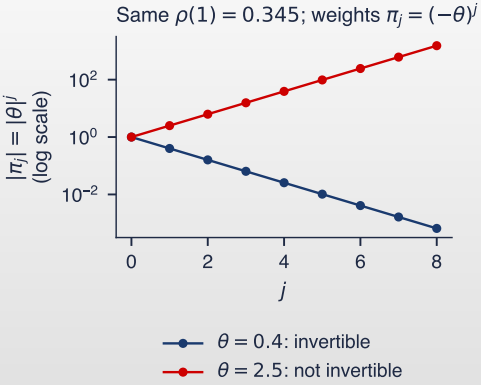
$$\gamma(0) = \sigma^2(1+\theta^2), \quad \gamma(1) = \theta\sigma^2, \quad \rho(1) = \frac{\theta}{1+\theta^2}, \quad \rho(h) = 0\ (h \geq 2)$$

  - $|\rho(1)| \leq 0.5$ for every θ
- **ARMA(1,1)** $X_t = \phi X_{t-1} + \varepsilon_t + \theta\varepsilon_{t-1}$

$$\psi_j = \phi^{j-1}(\phi+\theta), \quad \gamma(0) = \sigma^2\frac{1+2\phi\theta+\theta^2}{1-\phi^2}, \quad \rho(1) = \frac{(1+\phi\theta)(\phi+\theta)}{1+2\phi\theta+\theta^2}, \quad \rho(h) = \phi\rho(h-1)$$

  - ψ_j for $j \geq 1$; after lag 1 the ACF decays like that of an AR(1)

Prerequisites for Today (7/17): invertibility of an MA(1)



- **Two MA(1) with the same ACF**: θ with σ^2 and $1/\theta$ with $\theta^2\sigma^2$
  - $\rho(1) = \theta/(1+\theta^2)$ is the same for θ and $1/\theta$: chart, $\theta = 0.4$ and $\theta = 2.5$, $\rho(1) = 0.345$
  - the data cannot tell them apart: by convention, the invertible one is reported
- **The weights** $\pi_j = (-\theta)^j$ of $\varepsilon_t = \sum_j \pi_j X_{t-j}$
  - $|\theta| < 1$: the weights fade, old values matter little: the shocks can be recovered
  - $|\theta| > 1$: the weights explode (log scale): no sensible recovery of the shocks
  - every finite MA is stationary, whatever θ

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Prerequisites for Today (8/17): identification with the ACF and the PACF

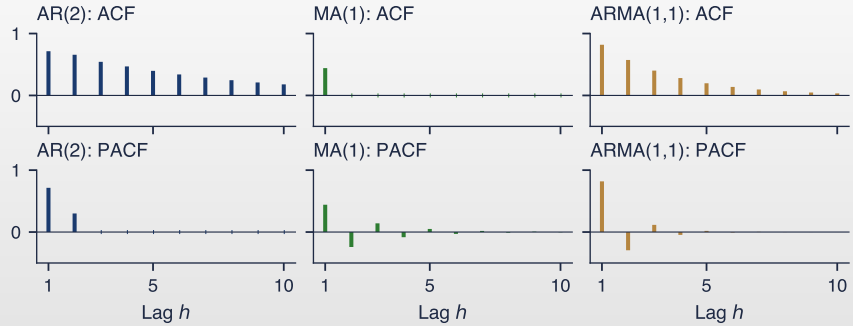
- **PACF** (partial autocorrelation function): ϕ_{hh}, the last coefficient of an AR(h) fitted to the series

$$X_t = \phi_{h\mathbf{1}} X_{t-\mathbf{1}} + \cdots + \phi_{hh} X_{t-h} + u_t$$

  - ϕ_{hh}: the link between X_t and X_{t-h} after removing the effect of the values in between
  - $\phi_{\mathbf{11}} = \rho(1)$; for an AR($p$), $\phi_{hh} = 0$ for $h > p$

- **Identification** rules
  - AR(p): the PACF cuts off after lag p, the ACF decays
  - MA(q): the ACF cuts off after lag q, the PACF decays
  - ARMA(p, q): both decay; the sample values are compared with the band $\pm 1.96/\sqrt{T}$

Prerequisites for Today (9/17): theoretical ACF and PACF



- AR(2) with $\phi = (0.5, 0.3)$: the ACF decays, the PACF has two bars ($\phi_{22} = 0.3$), then 0
- MA(1) with $\theta = 0.6$: one ACF bar ($\rho(1) = 0.44$), an alternating, decaying PACF
- ARMA(1,1) with $\phi = 0.7$, $\theta = 0.4$: neither cuts off; both decay

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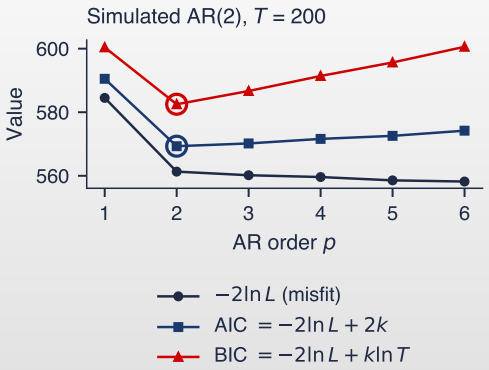
Prerequisites for Today (10/17): estimation

- **Yule–Walker** for AR(2): replace $\rho(1), \rho(2)$ by their sample values $\hat{\rho}_1, \hat{\rho}_2$ and solve

$$\hat{\phi}_1 = \frac{\hat{\rho}_1(1-\hat{\rho}_2)}{1-\hat{\rho}_1^2}, \quad \hat{\phi}_2 = \frac{\hat{\rho}_2-\hat{\rho}_1^2}{1-\hat{\rho}_1^2}, \quad \hat{\sigma}^2 = \hat{\gamma}(0)(1-\hat{\phi}_1\hat{\rho}_1-\hat{\phi}_2\hat{\rho}_2)$$

  - $\hat{\sigma}^2$: the estimated variance of the shocks; $\hat{\gamma}(0)$: the sample variance
- **Maximum likelihood** (MLE): the parameters that make the observed series most probable
  - the default of `ARIMA(x, order=(p, 0, q))` in `statsmodels`; `const` is the mean μ
- **Reading the output**
  - t-ratio $t = \hat{\phi}/\mathrm{SE}(\hat{\phi})$, SE: the standard error; $|t| > 1.96$: significant at 5%
  - $R^2 = 1 - \hat{\sigma}^2/\hat{\gamma}(0)$: the share of the variance explained by the model; for an AR(1), $R^2 \approx \hat{\phi}^2$

Prerequisites for Today (11/17): information criteria



- **AIC** (Akaike) and **BIC** (Bayesian, Schwarz): fit plus a penalty for each parameter

$$\text{AIC} = -2\ln L + 2k, \qquad \text{BIC} = -2\ln L + k\ln T$$

  - $\ln L$: the maximised log-likelihood; $-2\ln L$ measures the misfit and falls as parameters are added
  - k: the number of parameters (ϕ, θ, μ, σ^2); T: the number of observations
  - the smallest value wins; $\ln T > 2$ for $T \geq 8$, so BIC prefers smaller models
- **Chart**: AR(p) fitted to a simulated AR(2)
  - $-2\ln L$ keeps falling, but by less than the penalty after $p = 2$: both criteria choose $p = 2$ (circles)

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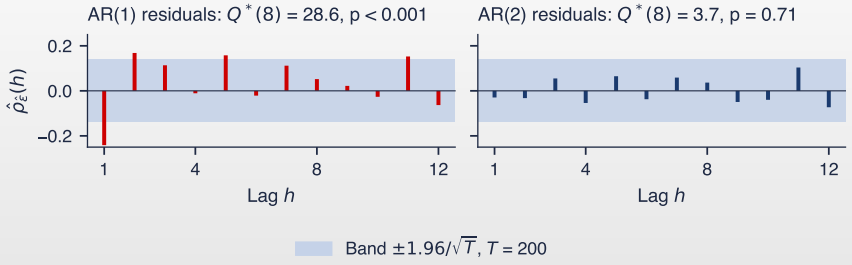
Prerequisites for Today (12/17): residual diagnostics

- **Residuals** $\hat{\varepsilon}_t = X_t - \hat{X}_{t|t-1}$: the one-step errors of the fitted model
  - $\hat{X}_{t|t-1}$: the value the model predicts for t with data up to $t-1$; a good model leaves white-noise residuals
- **Ljung–Box on residuals**: H_0: the residuals are white noise up to lag m

$$Q^*(m) = T(T+2)\sum_{h=1}^{m} \frac{\hat{\rho}_{\hat{\varepsilon}}(h)^2}{T-h} \approx \chi^2(m-p-q)$$

  - $\hat{\rho}_{\hat{\varepsilon}}(h)$: the sample ACF of the residuals; m: the number of lags tested
  - $m-p-q$ degrees of freedom: each estimated ARMA coefficient uses up one
  - reject if $Q^*(m) > \chi^2_{0.95}(m-p-q)$: the model left some autocorrelation
  - 5% critical values: $\chi^2_{0.95}(5) = 11.07$, $\chi^2_{0.95}(7) = 14.07$, $\chi^2_{0.95}(8) = 15.51$, $\chi^2_{0.95}(9) = 16.92$, $\chi^2_{0.95}(10) = 18.31$

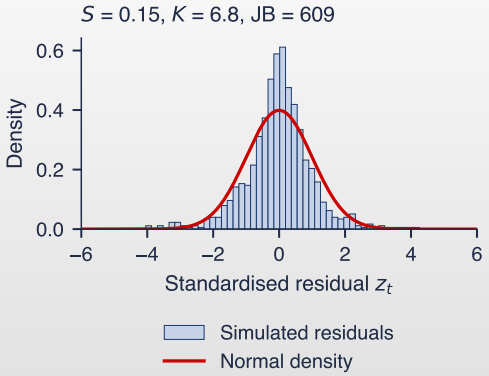
Prerequisites for Today (13/17): residuals of a model that is too small



- The simulated AR(2) of the previous charts, $T = 200$, fitted by an AR(1) and by an AR(2)
- Residual ACF and Ljung–Box with $8 - p$ degrees of freedom
  - AR(1): bars outside the band, $Q^*(8) = 28.6$, $p < 0.001$: the missing ϕ_2 stays in the residuals
  - AR(2): all bars inside, $Q^*(8) = 3.7$, $p = 0.71$: white-noise residuals

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Prerequisites for Today (14/17): skewness, kurtosis and Jarque–Bera



- **Skewness** S and **kurtosis** K of the standardised residuals $z_t = (\hat{\varepsilon}_t - \bar{\hat{\varepsilon}})/\hat{\sigma}$

$$S = \frac{1}{T}\sum_{t=1}^{T} z_t^3, \qquad K = \frac{1}{T}\sum_{t=1}^{T} z_t^4$$

  - Normal distribution: $S = 0$, $K = 3$; $K > 3$: heavy tails, more extreme values

- **Jarque–Bera** (JB): H_0: the residuals are Normal

$$JB = \frac{T}{6}\left(S^2 + \frac{(K-3)^2}{4}\right) \approx \chi^2(2)$$

  - reject at 5% if $JB > 5.99$; chart: $T = 1\,000$, $K = 6.8$ gives $JB = 609$
  - with Normal intervals, the extreme values are more frequent than the 5% they promise

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Prerequisites for Today (15/17): forecasts

- **Forecasts**: replace the future shocks by 0 and the future values by their forecasts
  - $\hat{X}_{T+h}$: the forecast made at T for h periods ahead; X_T: the last observation
- **AR(1)**: the forecast returns geometrically to the mean; **MA(q)**: the mean after q steps

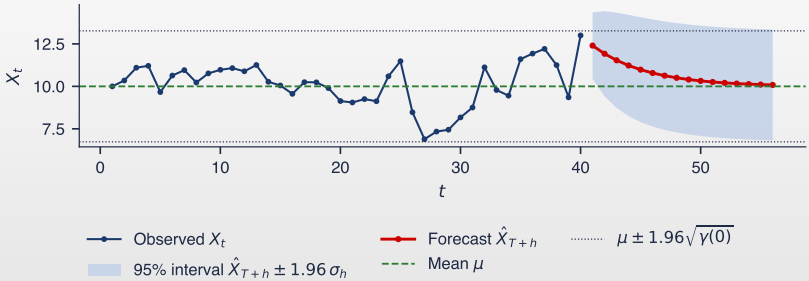
$$\hat{X}_{T+h} = \mu + \phi^h(X_T - \mu), \qquad \text{MA}(q)\text{: } \hat{X}_{T+h} = \mu \quad (h > q)$$

- **Forecast error variance**: the shocks that arrive between $T+1$ and $T+h$

$$\sigma_h^2 = \sigma^2(1 + \psi_1^2 + \cdots + \psi_{h-1}^2)$$

  - for a stationary model $\sigma_h^2 \to \gamma(0)$: the interval stops widening
  - 95% interval $\hat{X}_{T+h} \pm 1.96\,\sigma_h$; 1.96: the 97.5% quantile of the standard Normal distribution

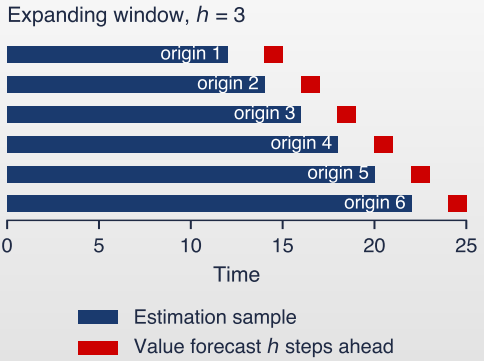
Prerequisites for Today (16/17): AR(1) forecasts and intervals



- A simulated AR(1) with $\mu = 10$, $\phi = 0.8$, $\sigma = 1$; the last value is $X_T = 13$
- Forecasts and their intervals
  - $\hat{X}_{T+\mathbf{1}} = 10 + 0.8 \cdot 3 = 12.4$, $\hat{X}_{T+\mathbf{2}} = 11.92$: the gap to the mean shrinks by 20% each period
  - $\sigma_{\mathbf{1}} = 1$, $\sigma_{\mathbf{2}} = \sqrt{1 + 0.64} = 1.28$; the band tends to $\mu \pm 1.96\sqrt{\gamma(0)} = 10 \pm 3.27$

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Prerequisites for Today (17/17): measuring forecast accuracy



- **Forecast errors** $e_i = X_{o_i+h} - \hat{X}_{o_i+h}$, $i = 1, \ldots, n$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_i^2}, \qquad \text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |e_i|$$

  - o_i: the forecast origin, the last date used; n: the number of forecasts
  - RMSE (root mean squared error) penalises large errors more; MAE (mean absolute error): the average size (Chapter 0)

- **Expanding window**: at each origin, estimate on all data up to o_i, forecast h steps ahead
  - only past data enter each forecast: an honest, out-of-sample comparison
  - benchmarks: the historical mean and the last value (the naive forecast)

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A1: an AR(1) process [Solved]

Task

- $X_t = 1 + 0.6X_{t-\mathbf{1}} + \varepsilon_t$, $\varepsilon_t \sim \text{WN}(0, 0.64)$, Gaussian.
- 1. Check stationarity and compute the mean and $\gamma(0)$.
- 2. Compute $\rho(2)$, $\rho(3)$ and the half-life of a shock.
- 3. With $X_T = 4$, forecast $X_{T+\mathbf{1}}$ and $X_{T+\mathbf{2}}$.
- 4. Compute the 95% intervals for both forecasts.
- Report: eight numbers and one sentence on mean reversion.

Solution

- 1. $|0.6| < 1$: stationary; $\mu = 1/0.4 = 2.5$; $\gamma(0) = 0.64/(1 - 0.36) = 1.00$
- 2. $\rho(2) = 0.6^{\mathbf{2}} = 0.36$, $\rho(3) = 0.216$; half-life $\ln 0.5 / \ln 0.6 = 1.36$ periods
- 3. $\hat{X}_{T+\mathbf{1}} = 2.5 + 0.6 \cdot 1.5 = 3.40$; $\hat{X}_{T+\mathbf{2}} = 2.5 + 0.36 \cdot 1.5 = 3.04$
- 4. $\sigma_{\mathbf{1}} = 0.8$: $[1.832, 4.968]$; $\sigma^{\mathbf{2}}_{\mathbf{2}} = 0.64(1 + 0.36) = 0.8704$, $\sigma_{\mathbf{2}} = 0.933$: $[1.21, 4.87]$
- The forecasts move from 4 back towards 2.5, and the intervals widen towards ± 1.96.

Back Exercise list

A2: an AR(2) process [Proposed]

Solution discussed in class

Task

- $X_t = 0.7X_{t-\mathbf{1}} - 0.1X_{t-\mathbf{2}} + \varepsilon_t$, $\varepsilon_t \sim \mathrm{WN}(0, 1)$. Model: A1.
- 1. Write the model with the lag operator and factorise $\phi(z)$.
- 2. Find the roots and decide whether the process is stationary; check the three triangle conditions.
- 3. Compute $\rho(1)$, $\rho(2)$ and $\rho(3)$ from the Yule–Walker equations.
- 4. Compute $\psi_{\mathbf{1}}$, $\psi_{\mathbf{2}}$ and $\psi_{\mathbf{3}}$.
- Report: the factorisation, two roots and six numbers.

A3: an MA(1) that is not invertible [Solved]

Task

- $X_t = \varepsilon_t + 2\varepsilon_{t-\mathbf{1}}$, $\varepsilon_t \sim \mathrm{WN}(0, 1)$.
- 1. Compute $\gamma(0)$, $\gamma(1)$ and $\rho(1)$.
- 2. Say whether the process is stationary and whether it is invertible.
- 3. Find the invertible MA(1) with the same autocovariances.
- 4. Write the first four weights of its AR(∞) form and the forecast $\hat{X}_{T+\mathbf{2}}$.
- Report: five numbers, two verdicts and the new model.

Solution

- 1. $\gamma(0) = 1 + 4 = 5$, $\gamma(1) = 2 \cdot 1 = 2$, $\rho(1) = 2/5 = 0.4$
- 2. Stationary (every finite MA is); not invertible: the root of $1 + 2z$ is -0.5, inside the circle
- 3. $\theta^* = 1/2$, $\sigma^{*\mathbf{2}} = 2^{\mathbf{2}} \cdot 1 = 4$: $\gamma(0) = 4 \cdot 1.25 = 5$, $\gamma(1) = 0.5 \cdot 4 = 2$
- 4. $u_t = X_t - 0.5X_{t-\mathbf{1}} + 0.25X_{t-\mathbf{2}} - 0.125X_{t-\mathbf{3}} + \ldots$; $\hat{X}_{T+\mathbf{2}} = \mu = 0$ (beyond $q = 1$)

Back Exercise list

A4: an ARMA(1,1) process [Proposed]

Solution discussed in class

Task

- $X_t = 0.5X_{t-\mathbf{1}} + \varepsilon_t + 0.3\varepsilon_{t-\mathbf{1}}$, $\varepsilon_t \sim \mathrm{WN}(0, 1)$. Model: A1, A3.
- 1. Check stationarity and invertibility.
- 2. Compute $\psi_{\mathbf{1}}$, $\psi_{\mathbf{2}}$ and $\psi_{\mathbf{3}}$.
- 3. Compute $\gamma(0)$, $\rho(1)$ and $\rho(2)$.
- 4. Show that $X_t = 0.5X_{t-\mathbf{1}} + \varepsilon_t - 0.5\varepsilon_{t-\mathbf{1}}$ is white noise, and say what an estimation program would report for it.
- Report: two verdicts, six numbers and one sentence.

A5: Yule–Walker and information criteria [Solved]

Task

- A stationary series has $T = 100$, $\hat{\gamma}(0) = 4$, $\hat{\rho}(1) = 0.75$, $\hat{\rho}(2) = 0.65$.
- 1. Estimate an AR(2) by Yule–Walker: $\hat{\phi}_1$, $\hat{\phi}_2$, $\hat{\sigma}^2$.
- 2. Check that the estimated model is stationary.
- 3. Maximum likelihood gives $\ln L = -214.6$ for AR(1) ($k = 3$) and -210.9 for AR(2) ($k = 4$); compute AIC and BIC.
- 4. Choose a model with each criterion.
- Report: three estimates, four criteria and a choice.

Solution

- 1. $1 - 0.75^2 = 0.4375$; $\hat{\phi}_1 = 0.75 \cdot 0.35/0.4375 = 0.6$; $\hat{\phi}_2 = (0.65 - 0.5625)/0.4375 = 0.2$
- $\hat{\sigma}^2 = 4\,(1 - 0.6 \cdot 0.75 - 0.2 \cdot 0.65) = 1.68$
- 2. $\hat{\phi}_1 + \hat{\phi}_2 = 0.8 < 1$, $\hat{\phi}_2 - \hat{\phi}_1 = -0.4 < 1$, $|\hat{\phi}_2| < 1$: stationary
- 3. $\ln 100 = 4.605$; AR(1): AIC $= 435.2$, BIC $= 443.02$; AR(2): AIC $= 429.8$, BIC $= 440.22$
- 4. Both criteria choose AR(2): the gain in $\ln L$ (3.7) exceeds both penalties per parameter (1 and $\ln(100)/2$).

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A6: four candidates and the Ljung–Box test [Proposed]

Solution discussed in class

Task

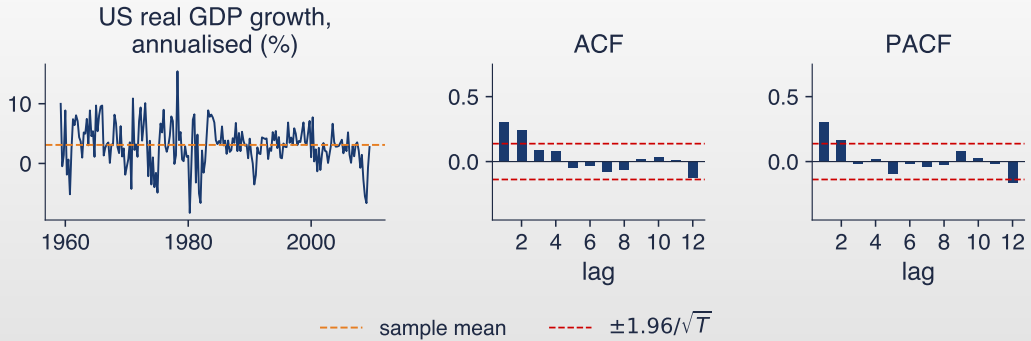
- $T = 120$; MLE gives $\ln L = -250.3$ (AR(1), $k = 3$), -247.1 (AR(2), $k = 4$), -247.6 (ARMA(1,1), $k = 4$), -246.9 (ARMA(2,1), $k = 5$). Model: A5.
- 1. Compute AIC and BIC for the four models.
- 2. Choose a model with each criterion.
- 3. The residuals of AR(2) give $Q^*(10) = 14.2$; find the degrees of freedom and decide at 5%.
- 4. Explain what changes if one uses 10 degrees of freedom.
- Report: a table of eight numbers, two choices and one decision.

B1: US real GDP growth [Solved]

- **Question**: which ARMA model describes US quarterly real GDP growth, and how persistent are growth shocks?
- **Inputs**: statsmodels `macrodata`, real GDP, 1959Q2–2009Q3; $y_t = 400\,\Delta \ln Y_t$ (annualised %)
- **Tasks**
  1. Plot y_t and its ACF and PACF up to lag 12, and propose candidate models.
  2. Estimate white noise, AR(1), AR(2), MA(2) and ARMA(1,1) by maximum likelihood, and tabulate AIC and BIC.
  3. Test the residuals of the BIC model with Ljung–Box $Q^*(8)$ on $8 - p - q$ degrees of freedom, and with Jarque–Bera.
  4. Interpretation: how many quarters does it take for half of a growth shock to fade?
- **Report**: the chart, a table of ten criteria, two test results and one sentence
- Notebook: section B1

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B1: solution (1/2) [Solved]



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B1: solution (2/2) [Solved]

	WN	AR(1)	AR(2)	MA(2)	ARMA(1,1)
AIC	1084.56	1066.98	**1063.70**	1065.59	1065.02
BIC	1091.17	**1076.91**	1076.93	1078.83	1078.25

- $T = 202$, ACF 0.30, 0.24, 0.09 and PACF 0.30, 0.16, -0.02 (band ± 0.14): AR(1) or AR(2)
- BIC: AR(1), $\hat{\mu} = 3.12$, $\hat{\phi} = 0.306$ (0.060); AIC: AR(2), $\hat{\phi}_2 = 0.163$ (0.062); the BIC values differ by 0.02
- AR(1) residuals: $Q^*(8) = 11.08$ on 7 df, p = 0.135; Jarque–Bera 16.0, $p < 0.001$
- Interpretation: half-life $\ln 0.5 / \ln 0.306 = 0.59$ quarters: growth shocks are short-lived; output levels, not growth rates, carry the persistence

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B2: Romanian real GDP growth [Proposed]

- **Question**: does Romanian quarterly real GDP growth have any ARMA structure?
- **Inputs**: Eurostat, chain-linked volumes, seasonally and calendar adjusted, 2000Q1–2026Q2; $y_t = 100\,\Delta \ln Y_t$; model: B1
- **Tasks**
  1. Repeat the four steps of B1 for y_t (`ro_gdp_growth('qoq')`).
  2. Run Ljung–Box $Q^*(8)$ on the series itself.
  3. Repeat the analysis without the four quarters of 2020, and compare.
  4. Interpretation: why is the best forecast of next quarter's growth simply the mean?
- **Report**: the chart, a table of ten criteria, the tests and two sentences
- Notebook: section B2

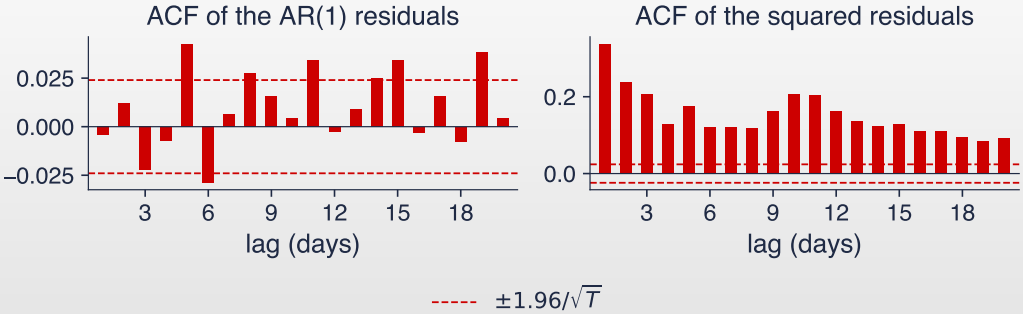
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B3: BET daily returns [Solved]

- **Question**: is the first-order autocorrelation of BET returns useful for forecasting?
- **Inputs**: BET closes since 2000 (EODHD); $r_t = 100\,\Delta \ln P_t$, $T = 6\,670$, last day 18 September 2026
- **Tasks**
  1. Estimate an AR(1) by maximum likelihood and report $\hat{\phi}$, its standard error and t-ratio, and R^2.
  2. Test the residuals with Ljung–Box $Q^*(10)$ on 9 degrees of freedom, and the squared residuals with $Q^*(10)$.
  3. Compute the kurtosis and the Jarque–Bera statistic of the residuals.
  4. Forecast tomorrow's return with its 95% interval.
  5. Interpretation: is the AR(1) coefficient statistically significant, and is it economically useful?
- **Report**: six numbers, the chart, a forecast with its interval and two sentences
- Notebook: section B3

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B3: solution (1/2) [Solved]



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B3: solution (2/2) [Solved]

- $\hat{\phi} = 0.109$ (se 0.0052), $t = 20.9$; $R^2 = 1.2\%$; $\hat{\sigma} = 1.390\%$
- Residuals: $Q^*(10) = 29.7$, p < 0.001 (some correlation remains); squares: $Q^*(10) = 2496$; kurtosis 14.0, Jarque–Bera 34128
- Last return 2.17%: forecast 0.294%, 95% interval $[-2.43, 3.02]$ (Normal, too narrow in the tails)
- Interpretation: highly significant, but it explains about 1% of the variance and the forecast is tiny relative to the interval; the large Q^* of the squares points to volatility models (Chapter 5)

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B4: EUR/RON daily returns [Proposed]

- **Question**: does the EUR/RON exchange rate behave like the BET?
- **Inputs**: BNR reference rate since July 2005, $T = 5\,352$ returns; model: B3
- **Tasks**
  1. Repeat the four steps of B3 (`b3_returns('eurron')`).
  2. Compare $\hat{\phi}$, R^2 and the kurtosis with those of the BET.
  3. Interpretation: why can a managed exchange rate show a larger first-order autocorrelation than a stock index?
- **Report**: a table (two series, six numbers each) and two sentences
- Notebook: section B4

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C1: does an AR model beat simple forecasts of inflation? [Proposed]

- **Question**: is an AR(2) better than the historical mean and the last value at forecasting Romanian 12-month inflation one year ahead?
- **Inputs**: HICP inflation since 2005 (Eurostat); forecast origins every month from January 2015; models: B1, A1
- **Tasks**
  1. At each origin, estimate an AR(2) on all data up to that month and forecast 12 months ahead.
  2. Compute the same forecasts from the historical mean and from the last observed value.
  3. Compare RMSE and MAE of the three methods, over the full period and before July 2021.
  4. Interpretation: why do all three methods fail in 2022?
- **Report**: a table of RMSE and MAE, the chart and a plan for a project
- Notebook: section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant about the ARMA models of this seminar. The answer:
- `(a) The process X(t) = 0.6 X(t-1) + 0.5 X(t-2) + e(t) is stationary because both coefficients are below 1.`
- `(b) An MA(1) with theta = 2 is not stationary, because |theta| > 1.`
- `(c) The AR(1) for BET returns has t = 21, so yesterday's return explains most of today's return.`
- `(d) For the residuals of an ARMA(1,1), Ljung-Box Q(8) is compared with the chi-square distribution with 8 degrees of freedom.`
- `(e) The forecast of an MA(2) three steps ahead equals the mean of the series.`
- `(f) In an AR(1), X(t) = c + phi X(t-1) + e(t), the mean of the series is c.`
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of six verdicts with one line of justification each.

Key takeaways

- Stationarity is read from the roots of $\phi(z)$, invertibility from the roots of $\theta(z)$, never from single coefficients
- The mean of an AR model is $c/\phi(1)$, not the intercept c
- Box–Jenkins: ACF and PACF propose, AIC and BIC choose, residual tests check, forecasts are compared with simple benchmarks
- Ljung–Box on residuals uses $m - p - q$ degrees of freedom
- Statistically significant is not economically large: compare $\hat{\phi}^2$ with the forecast interval

After the seminar

- Lecture 2 derives today's formulas: roots, invertibility, ψ weights, estimators, information criteria, forecast intervals, the Box–Jenkins case of Romanian GDP
- C1 can grow into a team project: ARMA forecasts of inflation in Romania and the region against simple benchmarks
- Reading: Huang and Petukhina (2022), Ch. 3–4; Hyndman and Athanasopoulos (2021), Ch. 9; Brockwell and Davis (2016), Ch. 3

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